# Cesare Huang

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## Research Interests

Post-quantum cryptography; computational number theory; high-performance computing; low-level program optimization.

## Research Experience

**Institute of Information Science, Academia Sinica**, Research Assistant — Aug 2024 – Present

- Develop high-performance, constant-time AArch64/NEON implementations of number-theoretic algorithms, with a focus on SafeGCD-based modular inversion.
- Verify low-level cryptographic implementations with CryptoLine or HOL Light, depending on the task.
- Developed and formally verified the Armv8-A SafeGCD implementation used in *Accelerating and Verifying Constant-Time Modular Inversion* (EUROCRYPT 2026).

## Publications

**Accelerating and verifying constant-time modular inversion.** Daniel J. Bernstein, Han-Ting Chen, John R. Harrison, **Cesare Huang**, Gregory Maxwell, Bow-Yaw Wang, Pieter Wuille, and Bo-Yin Yang. — 2026

- *Advances in Cryptology – EUROCRYPT 2026, Part VII*, LNCS 16547, pp. 598–627.

## Manuscripts Under Review

**Strided Frobenius Additive FFT and its Application to HQC.** Ming-Shing Chen, Tun-You Chien, Chun-Ming Chiu, **Cesare Huang**, Han-Hsuan Lin, Chun-Tao Peng, and Bo-Yin Yang. — 2026

- *Cryptology ePrint Archive*, Paper 2026/1588. Preprint.

**Multiplying Not-So-Big Integers with FFTs**
*Finding the Crossover Point on Large Modern OoO ARM CPUs*
**Cesare Huang**, Daisy Meng-Wei Liu, David Shu-Yu Wu, Bow-Yaw Wang, and Bo-Yin Yang.

## Manuscripts in Preparation

**safeRNSgcd: Accelerating Falcon key generation with RNS in SafeGCD**
**Cesare Huang** and Bo-Yin Yang.

## Education

**National Taiwan University**, B.S. in Mathematics — Sept 2020 – June 2024

## Technical Skills

**Programming:** C, Python, AArch64/NEON assembly
**Verification and optimization:** CryptoLine, HOL Light, SLOTHY, qhasm

## Talks

**Accelerating and Verifying Constant-Time Modular Inversion** — May 2026
Conference presentation, EUROCRYPT 2026.

**SafeGCD (Bernstein–Yang Inversion)** — Oct 2025
Invited lecture, cryptographic engineering course, National Taiwan University.

**Number Theoretic Transform** Aug 2025
Invited lecture, Core Program, ZK Education.

## Teaching and Academic Service

**IEEE Transactions on Computers**, Peer Reviewer 2026

**Topics in Cryptanalysis**, Teaching Assistant Feb 2026 – June 2026

**Cryptographic Engineering**, Teaching Assistant Sep 2025 – Dec 2025

**Post-Quantum Cryptography**, Teaching Assistant Feb 2025 – June 2025

**Quantum Information and Computation**, Teaching Assistant Feb 2024 – June 2024

## Mentorship

**Academia Sinica, Summer Intern Program** July 2025 – Present

- Mentored two interns in AArch64 assembly programming and efficient big-integer multiplication, resulting in a manuscript currently under review.

## Open-Source Contributions

**SLOTHY Assembly Optimizer** link

- Contributed AArch64 instruction-parser definitions and microarchitectural parameters.

last update: Sept 2026